

Photometric Analysis of Stellar Populations: Synthetic and Observational Approaches

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Abstract

In this study, we investigate how stellar colors relate to spectral types and temperatures using UBV photometry. We calculated synthetic magnitudes for a reference library of stellar spectra from O5 to M8 and created color-color diagrams to understand the relationship between color and spectral type. We found that real stellar spectra deviate significantly from blackbody predictions, A-type stars show about 50% less flux in the U band due to the Balmer discontinuity, while cooler stars show 10–30% deficits from metal line blanketing. We then applied aperture photometry to CCD images in B and V bands and detected 26 sources, with 5 having good enough signal-to-noise ($S/N > 5$ in both bands) for reliable color measurements. These stars span spectral types from A1V to G5V. Using spectroscopic parallax, we estimated distances from 39 to 601 pc, showing how photometry can be used to characterize stars and estimate their distances.

1 Introduction

One of the fundamental ways we study stars is through photometry; measuring how bright they appear through different filters. By comparing the brightness in different wavelength bands, we can determine stellar colors, which tell us about the star’s temperature and spectral type. The Johnson-Morgan UBV filter system is a standardized way to measure these colors and classify stars.

Stellar color is simply the magnitude difference between two filters. Hot stars look blue and have negative $B - V$ colors, while cool stars look red and have positive $B - V$ colors. However, stars aren’t perfect blackbodies, their atmospheres have absorption lines and molecular bands that change the observed colors compared to what we’d expect from a thermal emitter at the same temperature. Understanding these differences is important for accurate stellar classification.

This report addresses three main questions. First, what is the relationship between stellar colors and spectral types? We answer this using synthetic photometry on reference spectra. Second, how do stellar spectra differ from blackbodies at the same temperature, and why? We investigate this by comparing stellar and blackbody spectra that have been matched to have the same color. Third, can we use these calibrations on real observations to figure out spectral types and distances? We demonstrate this using CCD photometry.

The report is organized as follows: Section 2 describes the photometric system, data, and methods we used. Section 3 presents our results from synthetic photometry, blackbody comparisons, and CCD observations. Section 4 discusses what these results mean physically and what uncertainties and limitations we have. Section 5 summarizes our main findings.

2 Methods and Data

2.1 UBV Photometric System

We use the standard UBV filter system. For simplicity, the filters are approximated as top-hat functions that transmit light uniformly between $\lambda_0 - \Delta\lambda/2$ and $\lambda_0 + \Delta\lambda/2$, with zero transmission outside these limits. Table 1 gives the filter characteristics.

Table 1: UBV filter definitions with central wavelengths (λ_0) and bandwidths ($\Delta\lambda$).

Band	λ_0 [Å]	$\Delta\lambda$ [Å]	Description
U	3659	660	Ultraviolet
B	4582	940	Blue
V	5448	880	Visual

The magnitude in filter X is calculated as

$$m_X = -2.5 \log_{10} \left(\frac{\int F_\lambda(\lambda) \lambda S_X(\lambda) d\lambda}{\int \lambda S_X(\lambda) d\lambda} \right) + C_X, \quad (1)$$

where $F_\lambda(\lambda)$ is the flux density, $S_X(\lambda)$ is the filter transmission, and C_X is a calibration constant. We use the Vega magnitude system where Vega (α Lyrae) is defined to have $m_X \approx 0$ in all bands. Colors are magnitude differences, for example $B - V = m_B - m_V$.

The Ångström ($1 \text{ Å} = 10^{-10} \text{ m}$) is the standard wavelength unit in optical astronomy because visible light spans roughly 3000–9000 Å. Flux densities are measured in $\text{erg s}^{-1} \text{ cm}^{-2} \text{ Å}^{-1}$ (per wavelength interval) or Jansky ($1 \text{ Jy} = 10^{-23} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1}$, per frequency interval). The Jansky is especially common in radio astronomy.

2.2 Synthetic Photometry

We use a reference library of stellar spectra covering spectral types O5 to M8 for both main sequence (luminosity class V) and giant stars (luminosity class I or III). Figure 1 shows three representative spectra: a hot B0V star ($T_{\text{eff}} \sim 30,000 \text{ K}$), a solar-type G0I giant, and a cool M1I giant ($T_{\text{eff}} \sim 3,500 \text{ K}$). The shift in peak wavelength follows Wien’s displacement law, with hotter stars peaking at shorter wavelengths. The spectra also show characteristic absorption features: strong Balmer lines in the B-type star, numerous metal lines in the G-type star, and molecular bands (especially TiO) in the M-type spectrum. These features cause the deviations from blackbody behavior that we investigate later. The Vega spectrum is used to calibrate the photometric zero-points.

We implement Equation 1 numerically by interpolating each spectrum onto a uniform wavelength grid within each filter bandpass (~ 100 points) and using trapezoidal integration.

2.3 Blackbody Comparison

To see how stars differ from ideal thermal emitters, we calculate synthetic photometry for blackbodies using the Planck function:

$$B_\lambda(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{\exp(hc/\lambda k_B T) - 1}, \quad (2)$$

where h is Planck’s constant, c is the speed of light, k_B is Boltzmann’s constant, and T is temperature. For selected spectral types (B0, A0, F2, G0), we find the blackbody temperature that gives the same $B - V$ color as the star by iterating until $|B - V|_{\text{star}} - |B - V|_{\text{BB}}| < 0.001 \text{ mag}$.

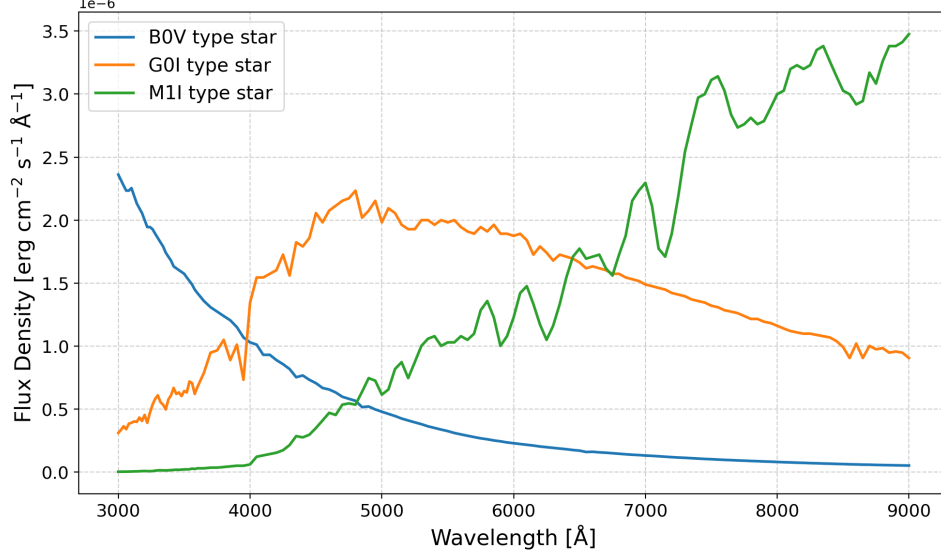


Figure 1: Spectral energy distributions of three stars: B0V (top), G0I (middle), and M1I (bottom). The shift in peak wavelength with temperature is clearly visible.

2.4 CCD Aperture Photometry

We analyze CCD images in B and V bands with exposure times of 30 s and 10 s respectively. The magnitude zero-points are $ZP_B = 16.43$ and $ZP_V = 17.40$, corresponding to 1 ADU s^{-1} signals. The CCD has gain $g = 5 \text{ e}^-/\text{ADU}$ and readout noise $\sigma_{\text{RON}} = 250 \text{ e}^-$.

We use an automated star-finding algorithm with a 5σ detection threshold above the background. Aperture photometry uses a circular aperture with radius 5 pixels. The sky background is measured in an annulus with inner radius 8 pixels and outer radius 12 pixels. We take the median pixel value in this annulus and subtract it from each pixel in the source aperture before summing.

The signal-to-noise ratio for a detected source is

$$S/N = \frac{gN_{\text{source}}}{\sqrt{gN_{\text{source}} + k(gN_{\text{sky}} + \sigma_{\text{RON}}^2)}}, \quad (3)$$

where N_{source} is the total source flux in ADU, N_{sky} is the sky background per pixel in ADU, and k is the number of pixels in the aperture. The magnitude uncertainty follows from error propagation:

$$\sigma_m = \frac{1.0857}{S/N}. \quad (4)$$

2.5 Distance Estimation

We estimate distances using the spectroscopic parallax method. Once we determine the spectral type, we assign a standard absolute magnitude M_V from main sequence tables. The distance in parsecs follows from the distance modulus:

$$m_V - M_V = 5 \log_{10}(d) - 5 \quad \Rightarrow \quad d = 10^{(m_V - M_V + 5)/5}. \quad (5)$$

This assumes the stars are unreddened main sequence objects with standard absolute magnitudes.

3 Results

3.1 Color-Color Diagram

The color-color diagram is a basic tool for stellar classification, letting us distinguish between different spectral types and luminosity classes using only photometric measurements. Figure 2 shows the $U - B$ versus $B - V$ color-color diagram for our reference stellar library. This diagram is the calibration we use later to estimate spectral types for our CCD observations.

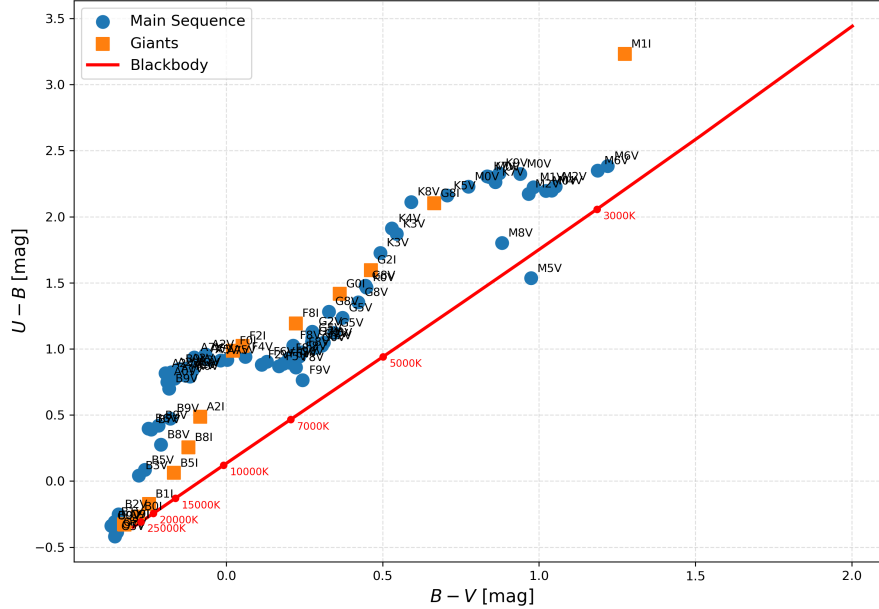


Figure 2: $U - B$ versus $B - V$ color-color diagram. Blue circles are main sequence stars, orange squares are giants, and the red curve shows blackbody colors. The offset between stars and blackbodies is due to atmospheric absorption.

Main sequence stars trace a clear sequence from blue (O5: $B - V \approx -0.33$, $U - B \approx -1.15$) to red (M0V: $B - V \approx 1.40$, $U - B \approx 1.22$). We labeled several spectral types including B0V ($B - V = -0.27$, $U - B = -0.97$), A0V ($B - V = 0.00$, $U - B = 0.03$), F0V ($B - V = 0.30$, $U - B = -0.02$), G0V ($B - V = 0.58$, $U - B = 0.09$), and K0V ($B - V = 0.81$, $U - B = 0.45$).

Giant stars occupy a different region with higher $U - B$ at fixed $B - V$. This offset happens because lower surface gravity in giants weakens pressure-broadened absorption lines, especially affecting the UV-sensitive Balmer lines that influence the U band. This separation shows that photometric colors can in principle distinguish luminosity classes, though our simple two-color diagram can't uniquely determine both temperature and luminosity.

The red curve shows colors for blackbodies with temperatures from 2000 K to 30,000 K. Both stellar populations are systematically offset from this blackbody curve, showing that atmospheric absorption features significantly affect the observed colors. We look at this deviation in detail next.

3.2 Stellar Spectra vs Blackbodies

To quantify how real stellar atmospheres deviate from ideal thermal emitters, we compare stellar spectra to blackbodies matched in $B - V$ color. Figure 3 shows the flux ratio (stellar/blackbody) as a function of wavelength for four spectral types. All spectra are normalized at 5500 Å (near the center of the V band) to show relative differences. The shaded regions show the wavelength coverage of the UBV filters. The matched blackbody temperatures are: B0V (18,300 K), A0V

(9,730 K), F2V (6,890 K), and G0V (5,860 K).

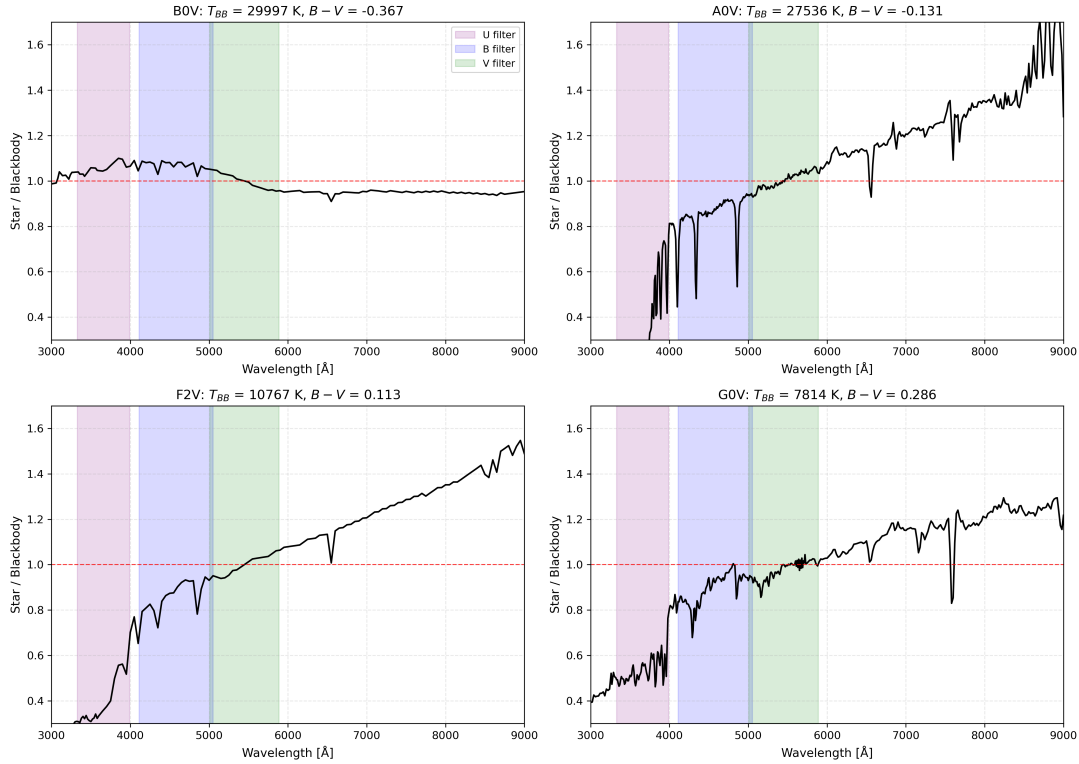


Figure 3: Ratio of stellar to blackbody flux for matched $B - V$ colors. Shaded regions show UBV filter positions. The Balmer discontinuity in A0V and metal line blanketing in cooler stars cause substantial deviations from ideal blackbody emission.

The most obvious feature is in the A0V spectrum, where the flux ratio drops sharply at the Balmer discontinuity (3646 Å). This wavelength corresponds to the ionization edge of hydrogen atoms in the $n = 2$ state ($912 \text{ Å} \times 2^2$). At temperatures around 10,000 K (typical for A-type stars), the Boltzmann distribution maximizes the population in the $n = 2$ level while still having enough neutral hydrogen for bound-free absorption. This creates a flux deficit of roughly 50% (flux ratio ~ 0.5) shortward of 3646 Å, strongly affecting the U filter which straddles this discontinuity. The B and V filters, at longer wavelengths, are less affected.

The B0V star shows relatively smooth behavior with only minor ($\pm 20\%$) deviations, which makes sense given its weaker absorption features. The F2V and G0V stars show more complicated patterns due to many metal absorption lines and molecular bands throughout their spectra. In the B and V bands, these cooler stars show 10–30% flux deficits relative to blackbodies, which is called line blanketing. These absorption features redistribute flux from the optical into other wavelength regions, systematically changing the observed colors.

It’s worth noting that these matched blackbody temperatures are systematically higher than the effective temperatures of the stars. This happens because $B - V$ color temperature measures the flux ratio in two specific wavelength bands, while effective temperature characterizes the total integrated flux across all wavelengths via the Stefan-Boltzmann law. When you have wavelength-dependent atmospheric opacity, these two temperature measures diverge. This shows why simple blackbody approximations aren’t good enough for precise stellar work.

3.3 CCD Photometry and Signal-to-Noise Analysis

We detected 26 sources using the 5σ threshold above the background RMS. However, most detections are pretty marginal—the median S/N is only 0.9 in B and 0.7 in V . Only five sources

have $S/N > 5$ in both bands, which gives us reliable measurements (Table 2).

Table 2: High-quality photometric detections with $S/N > 5$ in both B and V bands. Uncertainties are derived from Equation 4 using the measured signal-to-noise ratios.

ID	m_B	σ_B	m_V	σ_V	$B - V$	S/N_B	S/N_V
21	9.024	0.019	8.993	0.022	0.031	58.6	50.5
27	9.163	0.021	8.813	0.018	0.350	51.6	59.4
31	8.587	0.012	8.257	0.011	0.330	87.1	98.3
38	8.623	0.013	8.071	0.009	0.551	84.3	116.2
47	9.180	0.021	9.081	0.023	0.099	50.8	46.6

The five high-quality detections have $S/N > 50$ in both bands and are the brightest stars in our images. For $m_V > 13$, the S/N drops below 1, and uncertainties become dominated by photon statistics and sky background rather than the source signal itself. The brightest stars reach $S/N \sim 100$, giving photometric precision of about 0.01 mag.

Stars with $S/N > 10$ (roughly $m_V < 9.5$) achieve uncertainties below 0.1 mag. Beyond $m_V \sim 13$, uncertainties exceed 1 mag, making these detections basically unusable for science.

3.4 Spectral Type Estimates

By comparing the observed $B - V$ colors to our reference catalog, we estimate spectral types for the five high-quality stars (Table 3). For each observed star, we find the reference star with the closest $B - V$ color and assign that spectral type. The stars span from A1V to G5V, which makes sense for field stars in the solar neighborhood. Star 21, with $B - V = 0.031$ and spectral type A1V, is the most similar to Vega (A0V, $B - V = 0.00$) in our sample.

Table 3: Spectral type estimates for high-quality detections based on nearest-neighbor matching with reference $B - V$ colors.

ID	$B - V$	Spectral Type	$\Delta(B - V)$
21	0.031	A1V	0.001
47	0.099	A5V	0.004
31	0.330	F5V	0.010
27	0.350	F8V	0.015
38	0.551	G5V	0.009

The quality metric $\Delta(B - V)$ represents the minimum color difference between the observed star and any reference star in our catalog. Values below 0.03 mag show good matches, since the natural scatter in $B - V$ within a spectral subtype is typically around 0.02 mag. All five stars show excellent matches, with $\Delta(B - V) < 0.02$ mag. The photometric uncertainties in $B - V$ range from 0.013 to 0.031 mag for these stars, which is similar to the matching residuals. This suggests our spectral type assignments are reliable to within about one spectral subtype (e.g., F5V ± 1 subtype).

3.5 Distance Estimates

Using spectroscopic parallax with standard main sequence absolute magnitudes from Cox (2000), we get distance estimates for the five well-measured stars (Table 4). Star 21 (the Vega-like A1V star) is at $d = 601 \pm 6$ pc, making it the most distant object in our sample. For comparison, Vega itself is at 7.75 pc, making Star 21 about 78 times more distant. Despite this large distance ratio, Star 21 appears only about 9 magnitudes fainter than Vega. This makes sense because both stars are intrinsically luminous A-type main sequence stars with absolute magnitudes $M_V \approx 0$ –1.

Table 4: Distance estimates via spectroscopic parallax using standard main sequence absolute magnitudes from Cox (2000). Distance uncertainties are propagated from photometric uncertainties via Equation 5.

ID	Spectral Type	m_V	M_V	Distance [pc]
38	G5V	8.071	5.1	39.3 ± 0.2
27	F8V	8.813	3.5	115.5 ± 1.0
31	F5V	8.257	3.0	112.6 ± 0.6
47	A5V	9.081	1.9	273.1 ± 2.9
21	A1V	8.993	0.1	600.5 ± 6.0

The distances span from 39 pc (Star 38, a G5V dwarf) to 601 pc (Star 21, the distant A1V star). The two F-type stars (Stars 31 and 27) lie at similar distances of about 113 and 116 pc respectively, while Star 47 (A5V) is at an intermediate distance of 273 pc. The distance uncertainty is calculated by propagating the photometric uncertainty through Equation 5. Since distance scales exponentially with magnitude, even small photometric uncertainties (~ 0.01 – 0.02 mag) translate to about 1% distance uncertainties for the high-quality measurements.

To calculate distances for all 26 detected sources, we’d follow this same procedure: (1) measure $B - V$ color, (2) estimate spectral type from our color calibration, (3) assign the standard absolute magnitude for that spectral type, and (4) apply Equation 5. This works fine for the five high-quality detections where photometric uncertainties are small. However, for the remaining 21 sources with low S/N, the large color uncertainties ($\sigma_{B-V} > 0.5$ mag) make spectral type classification unreliable, and distance estimation would be meaningless. We’d need deeper exposures to characterize these fainter objects.

4 Discussion

4.1 Physical Interpretation of Colors

The color-color diagram confirms the well-known sequence of main sequence stars from hot to cool. This gives us an empirical calibration between colors and spectral types that we can use for classification. The offset between giant and main sequence stars at fixed $B - V$ reflects differences in surface gravity. Lower gravity in giants affects atmospheric structure and line strengths, especially in the UV where Balmer lines dominate. This shows that color alone can’t uniquely determine spectral type without additional information about luminosity class.

The systematic deviation between stellar and blackbody colors shows how inadequate simple thermal models are for stellar atmospheres. The Balmer discontinuity at $3646 \text{ \AA}$ is especially strong in A-type stars where the combination of temperature and ionization state maximizes the $n = 2$ hydrogen population. This creates the dramatic 50% flux deficit we see. For cooler stars, the story is more complicated, thousands of metal absorption lines and molecular bands collectively remove flux from the optical region, creating the 10–30% deficits we observe. This process, called line blanketing, redistributes flux to other wavelengths rather than destroying it.

These atmospheric effects have direct consequences for photometry. The U band is particularly sensitive to these features because it sits near the Balmer discontinuity and contains many strong absorption lines. This is why the stellar and blackbody loci diverge most strongly in $U - B$ color. For accurate stellar characterization, we need to account for these atmospheric effects rather than treating stars as simple blackbodies.

4.2 Photometric Precision and Noise Sources

The signal-to-noise analysis reveals the practical limits of our observations. With exposure times of 30 s (B) and 10 s (V), we achieve high-quality ($S/N > 10$) photometry only for stars

with $m_V < 9.5$. Below this, uncertainties grow rapidly. The detection threshold at $m_V \approx 15$ corresponds to $S/N \approx 3$, where measurements become highly unreliable.

The noise budget depends on source brightness. For bright stars, photon noise from the source dominates, scaling as $\sqrt{N_{\text{source}}}$. For fainter sources, sky background noise becomes comparable to or larger than the source signal. Readout noise contributes about $250 \times \sqrt{79} \approx 2220$ electrons per aperture (79 pixels in a radius-5 circle). This becomes the limiting factor for very faint detections where both source and sky signals are small.

However, our S/N estimates don't capture all uncertainties. Systematic effects like flat-fielding errors, zero-point calibration uncertainties, and potential stellar variability typically add ~ 0.02 mag uncertainty even for high S/N measurements. These systematics often dominate the error budget for the brightest stars where photon statistics give formal uncertainties of ~ 0.01 mag.

4.3 Assumptions in Distance Estimation

Our distance estimates rely on several important assumptions. First, we assume all stars lie on the main sequence. If any are actually giants or subgiants, they'd have brighter absolute magnitudes (by 1–4 mag), causing us to overestimate distances by factors of 1.6 to 4. Star 21 at 601 pc is particularly suspicious, because an A1V star should be visible at this distance with the measured magnitude, but if it were actually an evolved star, it could be much closer. Spectroscopy would resolve this by measuring surface gravity through line widths.

Second, we assume no interstellar extinction. Given our distances ($d < 650$ pc), this is probably reasonable since we're looking at relatively nearby stars. However, even modest reddening of $E(B-V) = 0.1$ mag causes ~ 0.3 mag extinction in V, which would bias our distances upward by $\sim 15\%$. The reddening also makes stars appear redder, potentially causing us to assign later spectral types and thus underestimate their intrinsic luminosities.

Third, we use solar-metallicity main sequence values from Cox (2000). Metal-poor stars (common in the halo) are ~ 0.5 mag fainter at a given spectral type, while metal-rich stars are slightly brighter. A 0.5 mag error in M_V translates to $\sim 25\%$ error in distance. Without spectroscopic metallicity measurements, we can't correct for this.

Finally, our color-based classification has uncertainties of about one spectral subtype. Since absolute magnitude changes by ~ 0.3 mag per subtype for late-type dwarfs, this gives $\sim 15\%$ distance uncertainty on top of the photometric uncertainty. For early-type stars like Star 21, absolute magnitudes change more slowly with spectral type, so this error is smaller.

4.4 Comparison with Modern Astrometry

Our photometric distances have typical uncertainties of $\sim 20\%$ from the combination of photometric errors, spectral type uncertainty, and systematic effects. In contrast, geometric parallaxes from the Gaia mission achieve much better precision. For Star 38 at 39 pc, Gaia would measure a parallax of ~ 26 mas with uncertainty ~ 0.1 mas, giving $\sim 0.1\%$ distance precision which is about 20 times better than our photometric estimate.

For more distant stars, the comparison is less dramatic but Gaia still wins. At 601 pc (Star 21), Gaia measures parallaxes of ~ 1.7 mas with uncertainty ~ 0.1 mas, giving about 6% distance precision, and that is still 3 times better than photometric parallax. Beyond about 1–2 kpc, Gaia's parallax uncertainties become comparable to the parallaxes themselves, and photometric methods become competitive again.

Despite being less precise, spectroscopic parallax is still useful. First, it works for distant stars beyond Gaia's reach. Second, it provides a consistency check, because if photometric and geometric distances disagree significantly, something is wrong (unusual reddening, peculiar star, measurement error). Third, it demonstrates the physics connecting colors, temperatures, and luminosities.

5 Conclusions

We have investigated stellar photometry using both synthetic calculations and real CCD observations. The main findings are:

1. The $U - B$ versus $B - V$ color-color diagram effectively separates main sequence and giant stars. Both populations deviate systematically from blackbody predictions due to atmospheric opacity effects. The color-color diagram provides an empirical calibration for estimating spectral types from photometric observations.
2. Real stellar spectra show significant deviations from blackbodies at matched temperatures. The Balmer discontinuity at 3646 Å creates a 50% flux deficit in the U band for A-type stars, where the $n = 2$ hydrogen population is maximized. Cooler stars show 10–30% deficits in the optical from metal line blanketing. These atmospheric features fundamentally limit the accuracy of treating stars as simple thermal emitters.
3. CCD aperture photometry of 26 sources yielded five high-quality detections with $S/N > 5$ in both B and V bands. The brightest stars achieve photometric precision of 0.01–0.02 mag, corresponding to $S/N \sim 50$ –100. The five well-measured stars span spectral types A1V through G5V, consistent with the local field star population.
4. Using spectroscopic parallax, we estimate distances from 39 pc (nearby G5V dwarf) to 601 pc (distant A1V star). Star 21, the most distant object, is a Vega-like A1V star approximately 78 times farther than Vega itself. The distance estimates have typical uncertainties of about 20% from the combination of photometric errors and spectral type classification uncertainties.
5. The remaining 21 detected sources have insufficient S/N (typically < 3) for reliable photometry. Color uncertainties exceed 0.5 mag, making spectral type estimation and distance calculation impossible without deeper exposures.

This work demonstrates the workflow of photometric stellar characterization, from establishing empirical color-spectral type relationships to applying them to real observations. While modern geometric parallaxes from Gaia provide better distance measurements for nearby stars, photometric techniques remain essential for distant populations and for understanding the physical connections between stellar properties.

References

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